Part I
Bayesian Modelling and Decision Theory

1 Bayesian Formulation

1.1 Frequentist Approach

Loss function $l(\theta, \delta)$, frequentist risk function:

$$R(\theta, \delta) = \text{function of } \theta = E[l(\theta, \delta(X))] = \int_{\mathcal{X}} l(\theta, \delta(x)) dP_X$$

1.2 Bayesian Statistics

θ unknown modelled as a r.v.
$(\Theta, \mathcal{A}, \pi)$ probability space for θ, where:

- $\mathcal{A}$: σ-field
- π: prior probability
- $X|\theta \sim P_\theta$ conditional distribution of X given θ.

Prior $\pi \implies (X, \theta) \sim P_\Theta \times \pi$

Example 1.1

Let $\pi(\theta|x)$ be the conditional distribution of θ given X, i.e., the posterior distribution.
Non-parametric bayesian stat:

$$f(x) = \sum_{j \in \mathbb{N}} c_j \varphi_j(x), \quad (c_j)_{j \in \mathbb{N}}$$

For example: $J \sim \text{Poisson}(\lambda)$; $\forall i \in (0, 2)$

$$f(x, \theta) = \sum_{j=1}^{n} \pi_J(j) d\pi(f_{\theta,j}|J = j)$$

Definition 1.2: Dominated Model

Model $\mathcal{P} = \{P_\theta, \theta \in \Theta\}$ for observations $X \in \mathcal{X}$ is a dominated model if $\exists \mu$ measure on $\mathcal{X}$ such that $\forall \theta \in \Theta$:

$$P_\theta \ll \mu \quad \text{(absolutely continuous)}$$

$f_\theta = \frac{dP_\theta}{d\mu}$ (density). Same μ for all θ ($\mu \neq 0$ in general).

Theorem 1.3: Bayes Theorem

If $P_\theta \ll \mu \quad \forall \theta \in \Theta$, and if π is a probability measure with density w.r.t. ν on Θ, then the

posterior has a density w.r.t. ν:

$$\pi(\theta|x) = \frac{f_\theta(x)\pi(\theta)}{\int_\Theta f_\theta(x)\pi(\theta)d\nu(\theta)}$$

Example 1.4: Poisson Distribution

$X_1, \ldots, X_n \sim \mathcal{P}(\theta)$ (Poisson θ), where $X_i \in \mathbb{N}$:

$$P_\theta(X = k) = e^{-\theta}\frac{\theta^k}{k!}$$

Prior $\pi_b(\theta) = be^{-b\theta}, \quad \theta > 0$ (density w.r.t. Lebesgue on $\mathbb{R}$).
$\pi_b = \mathcal{E}(b)$
Posterior has a density w.r.t. Lebesgue:

$$\pi(\theta|x_1, \ldots, x_n) = \frac{\prod_{i=1}^n P_\theta(X_i = x_i)be^{-b\theta}}{\int_0^\infty \prod_{i=1}^n P_{\tilde{\theta}}(X_i = x_i)be^{-b\tilde{\theta}}d\tilde{\theta}}$$

$$\pi(\theta|x_1, \ldots, x_n) \propto e^{-(n+b)\theta}\theta^{\sum x_i}$$

$$\equiv \Gamma\left(\sum_{i=1}^n x_i + 1, n + b\right)$$

Recall $\Gamma(a', b')$ has a density w.r.t. Lebesgue:

$$\pi_{a',b'}(y) = e^{-b'y}y^{a'-1}\frac{(b')^{a'}}{\Gamma(a')}\mathbb{I}_{y>0}$$

Example 1.5: Normal Distribution

$X_1, \ldots, X_n$ i.i.d. $\mathcal{N}(\theta, 1)$, $\theta \in \mathbb{R}$
$\mu =$ Lebesgue on $\mathbb{R}^n$
$\nu =$ Lebesgue on $\mathbb{R}$
Prior $\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$
Posterior density $\pi(\theta|X)$:

$$\propto \prod_{i=1}^n \exp\left(-\frac{(X_i - \theta)^2}{2}\right) \times \exp\left(-\frac{(\theta - \mu_0)^2}{2\tau_0^2}\right)$$

$$\propto \exp\left(-\frac{\theta^2}{2}\left(n + \frac{1}{\tau_0^2}\right) + \theta\left(n\overline{X}_n + \frac{\mu_0}{\tau_0^2}\right)\right)$$

$$\equiv \mathcal{N}\left(\frac{n\overline{X}_n + \frac{\mu_0}{\tau_0^2}}{n + \frac{1}{\tau_0^2}}, \frac{1}{n + \frac{1}{\tau_0^2}}\right)$$

Definition 1.6: Conjugate Family

If the prior π belongs to $\mathcal{S}$, a family of distributions such that the posterior $\pi(\cdot|x) \in \mathcal{S}$ a.s., then $\mathcal{S}$ is called a conjugate family of priors w.r.t. the statistical model we are using.

Example 1.7

$X_1, \dots, X_n \sim \mathcal{P}(\theta)$, $\theta > 0$
$\mathcal{S} := \left\{ \pi_{a,b}(\theta) = \frac{\theta^{a-1} e^{-b\theta} b^a}{\Gamma(a)} \right\} \equiv \Gamma$ distrib
Then $\forall n$: $\pi(\theta|x_1, \dots, x_n) \sim \Gamma\left(a + n\overline{x}_n, b + n\right) \in \mathcal{S}$.

If the model is dominated, posterior density w.r.t. ν:

$$\pi(\theta|x) = \frac{f_\theta(x)\pi(\theta)}{m_\pi(x)} \propto f_\theta(x)\pi(\theta)$$

$m_\pi(x)$ is the marginal density of the data (marginal likelihood).

1.3 Hierarchical Prior

$\pi = \pi_\lambda$ (e.g. $\Gamma(a, b)$ with $\lambda = (a, b)$).
λ is called a hyperparameter.

Example 1.8

$X_i \sim \mathcal{N}(\theta_i, 1)$ for $i = 1, \dots, n$
$\theta = (\theta_1, \dots, \theta_n)$
$\theta_i \sim \mathcal{N}(0, \tau^2)$
$X_i \in \mathbb{R}$
τ^2 is here a hyperparameter that has a lot of influence.

- Choice 1: fix τ^2 arbitrarily.
- Choice 2: take $\tau^2 \sim$ Cauchy (positive part). Fat tails, so that τ^2 is sufficiently large.

2 Bayesian Decision Theory

2.1 Loss and Risks

Model $\mathcal{P}_\theta, \theta \in \Theta$ for observations X.

Definition 2.1: Loss Function

A loss function $l : \Theta \times \mathcal{D} \to \mathbb{R}_+$, where $\mathcal{D}$ represents decisions δ.

Example 2.2: Loss Functions

1. If $\mathcal{D} = \Theta$:
   - Quadratic loss: $l(\theta, \delta) = \|\theta - \delta\|^2$
   - L_1 loss: $l(\theta, \delta) = \|\theta - \delta\|_1 = \sum_{j=1}^d |\theta_j - \delta_j|, \quad \theta \in \mathbb{R}^d$
2. If $\mathcal{D} = \{g(\theta), \theta \in \Theta\}$, $g : \Theta \to \mathbb{R}$:

- $l(\theta, \delta) = \|g(\theta) - \delta\|^2$
- $l(\theta, \delta) = \|g(\theta) - \delta\|_1$

3. Test $\mathcal{D} = \{0, 1\}$:
   - ex: $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$
   - $\Theta_0 \cap \Theta_1 = \emptyset$
   - $l(\theta, \delta) = 1$ if θ and δ disagree, 0 otherwise.
4. $X_1, \ldots, X_n \sim f_\theta$ density w.r.t. Lebesgue:
   - $l(\theta, \delta) = \mathrm{KL}(f_\theta, f_\delta)$ (with $\delta = \Theta$)

Definition 2.3: Risks

- **Posterior risk** associated with decision δ:

$$\rho(\pi, \delta|x) = \int_\Theta l(\theta, \delta)\pi(d\theta|x)$$

- **Integrated risk**:

$$r(\pi, \delta) = \int \rho(\pi, \delta|x) m_\pi(dx)$$

(where m_π is the marginal distribution of X).

Definition 2.4: Bayesian Estimator

$$\delta^\pi(x) := \mathrm{argmin}_\delta \rho(\pi, \delta|x) \quad \text{for all } x$$

Lemma 2.5

For any estimator δ, $r(\pi, \delta^\pi) \leq r(\pi, \delta)$

Proof

For all x, $\rho(\pi, \delta^\pi|x) \leq \rho(\pi, \delta(x)|x)$

$$\implies \int \rho(\pi, \delta^\pi|x) m_\pi(dx) \leq \int \rho(\pi, \delta(x)|x) m_\pi(dx)$$

□

Lemma 2.6

If the model is dominated by μ,

$$r(\pi, \delta) = \int_\Theta R(\theta, \delta)\pi(d\theta)$$

Proof (Fubini Theorem)

$$
\begin{aligned}
r(\pi, \delta) &= \int_{\mathcal{X}} \int_{\Theta} l(\theta, \delta(x)) \frac{f_\theta(x)}{m_\pi(x)} \pi(d\theta) m_\pi(x) d\mu(x) \\
&= \int_{\Theta} \left(\int_{\mathcal{X}} l(\theta, \delta(x)) f_\theta(x) d\mu(x) \right) \pi(d\theta) \\
R(\theta, \delta) &= E[l(\theta, \delta(X))]
\end{aligned}
$$

□

Example 2.7: Quadratic Loss

Quadratic loss, $X|\theta \sim P_\theta$, $\theta \in \Theta \subset \mathbb{R}^d$
$l(\theta, \delta) = \|\theta - \delta\|^2$
Prior $\theta \sim \pi$
Posterior: $\pi(d\theta|x) = \frac{f_\theta(x) d\pi(\theta)}{\int_\Theta f_\theta(x) d\pi(\theta)}$
$\delta \to \rho(\pi, \delta|x)$ is convex.

$$
\frac{\partial}{\partial \delta} \rho(\pi, \delta|x) = -2 \int_\Theta (\theta - \delta) \pi(d\theta|x) = 0
$$

$$
\implies \delta(x) = \delta^\pi(x) = \int_\Theta \theta d\pi(\theta|x)
$$

which is the posterior mean.

Example 2.8: 0-1 Loss

$\Theta = \{1, \ldots, K\}$
f_θ density
Prior: $\pi = (\pi_1, \ldots, \pi_K)$
Loss: $l(\theta, \delta) = 1$ if $\theta \neq \delta$, and $= 0$ if $\theta = \delta$.

$$
\begin{aligned}
\rho(\pi, \delta|x) &= \sum_{k=1}^{K} l(k, \delta) \pi(k|x) \\
&= \sum_{k=1}^{K} \mathbb{I}_{k \neq \delta} \pi(k|x) = 1 - \pi(\delta|x) \\
\delta^\pi(x) &= \text{argmin}_\delta (1 - \pi(\delta|x)) = \text{argmax}_k \pi(k|x)
\end{aligned}
$$

2.2 Admissibility

Definition 2.9: Admissibility

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Example 2.10: The MLE is not admissible for $d = 3$

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Proof

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2.3 Intrinsic Losses

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Example 2.11: Kullback–Leibler Divergence

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Proof (invariance of the KL divergence)

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2.4 The MAP Estimator in Dominated Models

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Definition 2.12: Maximum A Posteriori (MAP) Estimator

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Remark: **TO COMPLETE**
Exercise (MAP estimator, exponential model with Gamma prior): **TO COMPLETE**

2.5 Improper Priors

Definition 2.13: Improper Prior

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Example 2.14: Lebesgue Measure on $\mathbb{R}$

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TO COMPLETE: posterior distribution associated with an improper prior

Example 2.15: Flat Prior on a Normal Mean

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Exercise (normal model with prior $\pi(\mu, \sigma^2) = 1/\sigma^2$): **TO COMPLETE**

Example 2.16: Haldane Prior for a Bernoulli Model

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2.5.1 Jeffreys Prior

TO COMPLETE: why use improper priors, flatness and parametrization

Definition 2.17: Jeffreys Prior

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Proposition 2.18: Invariance under Reparametrization

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Proof

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Exercise (Jeffreys prior for $\mathcal{N}(\mu, 1)$): TO COMPLETE

2.6 Nuisance Parameters: Marginal Likelihoods

Definition 2.19: Marginal Likelihood and Profile Likelihood

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Example 2.20: Normal Model with Unknown Variance

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2.7 Bayesian Tests

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Example 2.21: Test in a Linear Regression

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TO COMPLETE: Bayes test under the 0-1 loss

Definition 2.22: Bayes Factor

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Remark (Bayes test and Bayes factor): TO COMPLETE

Example 2.23: Two Simple Hypotheses

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Example 2.24: Point Null Hypothesis

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Remark (marginal likelihoods m_0 and m_1): TO COMPLETE

Proposition 2.25: Bayes Test, General Case

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Remark (improper priors and the Bayes test): TO COMPLETE

Proof

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2.8 Model Selection

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Example 2.26: Variable Selection in Linear Regression

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Exercise (posterior distribution on the model γ): TO COMPLETE

Example 2.27: Number of Components in a Mixture

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2.9 Other Losses for Tests

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Exercise (Bayes estimator for this loss): TO COMPLETE

2.10 Credible Regions

Definition 2.28: Confidence Region (Frequentist Recall)

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Definition 2.29: Credible Region

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Lemma 2.30: Credible Regions are Confidence Regions on Average

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Proof

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Example 2.31: Equal-Tail Credible Interval

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Example 2.32: Credible Interval in the Normal Model

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Definition 2.33: Highest Posterior Density (HPD) Region

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Theorem 2.34: HPD Regions Minimize the Volume

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Proof

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Example 2.35: Computation of an HPD Region

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Part II
Consistency and Convergence Rate

3 Consistency

TO COMPLETE: reminder on consistency of an estimator in frequentist evaluation

3.1 Definition

Let $X_{1:n} \sim P_\theta^{(n)}$, $\theta \in \Theta$, (Θ, d) = metric Polish space.
π = proba prior and $\pi(\cdot \mid X^{(n)}) \equiv$ posterior distribution.

Definition 3.1: Posterior Consistency

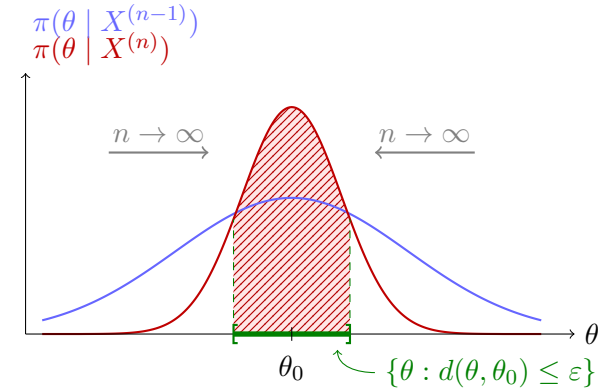
The posterior is a.s. *consistent at* θ_0 (w.r.t. the metric d) iff

$$\forall \varepsilon > 0, \quad \pi\big(\{\theta,\ d(\theta,\theta_0) \leq \varepsilon\} \,\big|\, X^{(n)}\big) \xrightarrow[n\to\infty]{} 1 \quad P_{\theta_0}^{(\infty)} \text{ a.s.}$$

TO COMPLETE: consistency in probability

Intuition

$X_{1:n} \sim P_{\theta_0}^{(n)}$



$$\text{▨}: \ \pi\big(\{\theta, d(\theta,\theta_0) \leq \varepsilon\} \,\big|\, X^{(n)}\big) \xrightarrow[n\to\infty]{P_{\theta_0}^{(\infty)} \text{ a.s.} \,/\, P_{\theta_0}^{(\infty)}} 1$$

Theorem 3.2

If (Θ, d) is a Polish space, the posterior is consistent a.s. at θ_0 (respect. in proba) iff

$$\pi(\cdot \mid X^{(n)}) \xrightarrow{\text{weakly}} \delta_{\theta_0} \quad \text{a.s. in } P_{\theta_0} \text{ (resp. in proba } P_{\theta_0}).$$

Proof

($\Rightarrow$) Let $\forall j \in \mathbb{N}^*$, $\varepsilon_j := \frac{1}{j}$.
So $\exists\, \Omega_j$ s.t. $P_{\theta_0}(\Omega_j) = 1$ and

$$\forall \omega \in \Omega_j, \quad \pi\big(B(\theta_0, \varepsilon_j) \mid X^{(n)}(\omega)\big) \longrightarrow 1.$$

Let $\Omega := \bigcap_{j\in\mathbb{N}^*} \Omega_j$, so $P_{\theta_0}(\Omega) = 1$ and

$$\forall j \in \mathbb{N}^*, \quad \pi\big(B(\theta_0, \varepsilon_j) \mid X^{(n)}(\omega)\big) \longrightarrow 1.$$

Let $\mathcal{O}$ be an open set of Θ:

- If $\theta_0 \in \mathcal{O}$, $\exists j \in \mathbb{N}^*$ s.t. $B(\theta_0, \varepsilon_j) \subset \mathcal{O}$, so

$$\pi\big(\mathcal{O} \mid X^{(n)}(\omega)\big) \geq \pi\big(B(\theta_0, \varepsilon_j) \mid X^{(n)}(\omega)\big) \longrightarrow 1$$

and $\delta_{\theta_0}(\mathcal{O}) = 1$ since $\theta_0 \in \mathcal{O}$. So $\pi(\mathcal{O} \mid X^{(n)}(\omega)) \xrightarrow[n\to\infty]{} \delta_{\theta_0}(\mathcal{O}) = 1$.

- If $\theta_0 \notin \mathcal{O}$, $\exists j \in \mathbb{N}^*$ s.t. $B(\theta_0, \varepsilon_j) \cap \mathcal{O} = \emptyset$, so

$$\begin{aligned}\pi\big(\mathcal{O} \mid X^{(n)}(\omega)\big) &= \pi\big(\mathcal{O} \cap B(\theta_0, \varepsilon_j)^c \mid X^{(n)}(\omega)\big) \\ &\leq \pi\big(B(\theta_0, \varepsilon_j)^c \mid X^{(n)}(\omega)\big) \longrightarrow 0.\end{aligned}$$

So $\pi(\mathcal{O} \mid X^{(n)}(\omega)) \longrightarrow 0 = \delta_{\theta_0}(\mathcal{O})$ (since $\theta_0 \notin \mathcal{O}$).

So $\forall \mathcal{O}$ open of Θ, $\pi(\mathcal{O} \mid X^{(n)}(\omega)) \longrightarrow \delta_{\theta_0}(\mathcal{O})$, so by Portmanteau lemma:

$$\pi\big(\cdot \mid X^{(n)}(\omega)\big) \longrightarrow \delta_{\theta_0}.$$

($\Leftarrow$) Let's suppose Ω s.t. $P_{\theta_0}(\Omega) = 1$ and $\forall \omega \in \Omega$, $\pi(\cdot \mid X^{(n)}(\omega)) \longrightarrow \delta_{\theta_0}$. Which is, by Portmanteau, equivalent to: $\forall \mathcal{O}$ open of Θ,

$$\pi\big(\mathcal{O} \mid X^{(n)}(\omega)\big) \xrightarrow[n\to\infty]{} \delta_{\theta_0}(\mathcal{O}).$$

So by taking $\forall \varepsilon$, $\mathcal{O} = B(\theta_0, \varepsilon)$, we obtain the conclusion. □

TO COMPLETE: PROOF OF CONSISTENCY IN $\mathbb{P}$

3.2 Schwartz Theorem

Theorem 3.3: Schwartz

Let Θ be a Polish space and $X^{(n)} = (X_1, \ldots, X_n)$,

$$X_i \mid \theta \overset{\text{iid}}{\sim} f_\theta, \quad \theta \in \Theta, \quad f_\theta \text{ density w.r.t. } \mu.$$

Let π be a proba on Θ. If

$$\forall \varepsilon > 0, \quad \pi\Big(\underbrace{\{\theta;\ \mathrm{KL}(\theta_0; \theta) < \varepsilon\}}\Big) > 0$$

we can view this as "$B_{\mathrm{KL}}(\theta_0, \varepsilon)$" (KL divergence is not a metric)

then $\forall A_n \subset \Theta$ (measurable), which verify either

i) $\exists\, 0 \leq \phi_n \leq 1$ a test function s.t. $\phi_n \longrightarrow 0$ $P_{\theta_0}^{\infty}$ a.s. and

$$\int_{A_n} \mathbb{E}_\theta^{(n)}(1 - \phi_n)\, d\pi(\theta) \leq e^{-Cn} \quad \forall n \geq n_0,\ C > 0,$$

or if

ii) $\exists\, C > 0$, $\pi(A_n) < e^{-Cn}$,

then

$$\pi(A_n \mid X^{(n)}) \longrightarrow 0 \quad P_{\theta_0}^{\infty} \text{ a.s.}$$

Remark:

- ϕ_n is a shortcut for $\phi_n(X^{(n)})$ where $\phi_n : \mathcal{X}^n \longrightarrow [0, 1]$.
- We can view the two conditions of (i) as controls on type I and type II errors.

Proof

Let $l_n(\theta) := \sum_{i=1}^n \ln f_\theta(X_i)$.

$$\pi(A_n \mid X^{(n)}) = \frac{\displaystyle\int_{A_n} \frac{f_\theta^{(n)}(X^{(n)})}{f_{\theta_0}^{(n)}(X^{(n)})}\, d\pi(\theta)}{\underbrace{\displaystyle\int_\Theta \frac{f_\theta^{(n)}(X^{(n)})}{f_{\theta_0}^{(n)}(X^{(n)})}\, d\pi(\theta)}_{=: D_n(X_1,\dots,X_n)}} = \frac{\displaystyle\int_{A_n} e^{l_n(\theta)-l_n(\theta_0)}\, d\pi(\theta)}{D_n}$$

Let's show that $\mathbb{1}_{D_n \le e^{-n\delta}} \xrightarrow{P_{\theta_0}^\infty \text{ a.s.}} 0$, $\forall \delta > 0$.

Let's prove $P_{\theta_0}^\infty\big(\lim D_n e^{n\delta} = +\infty\big) = 1$, $\forall \delta > 0$.

$$e^{n\delta} D_n = e^{n\delta} \int_\Theta e^{l_n(\theta)-l_n(\theta_0)}\, d\pi(\theta) \overset{①}{\ge} \int_{S^\varepsilon} e^{l_n(\theta)-l_n(\theta_0)+n\delta}\, d\pi(\theta)$$

where $S^\varepsilon := \{\theta \in \Theta : \mathrm{KL}(\theta_0, \theta) < \varepsilon\}$, $\pi(S^\varepsilon) > 0$ by hypothesis.

$$\liminf_{n\to\infty} \int_{S^\varepsilon} e^{l_n(\theta)-l_n(\theta_0)+n\delta}\, d\pi(\theta) \overset{②}{\ge} \int_{S^\varepsilon} \liminf_{n\to\infty} e^{l_n(\theta)-l_n(\theta_0)+n\delta}\, d\pi(\theta)$$

And since (by LLN)

$$\frac{l_n(\theta) - l_n(\theta_0)}{n} \xrightarrow{P_{\theta_0}^\infty \text{ a.s.}} \underbrace{-\mathrm{KL}(\theta_0, \theta)}_{> -\varepsilon \text{ if } \theta \in S^\varepsilon}$$

So by taking $\varepsilon < \delta$ and $\theta \in S^\varepsilon$, $\dfrac{l_n(\theta) - l_n(\theta_0)}{n} + \delta \longrightarrow -\mathrm{KL} + \delta > 0$,

$$\text{so} \quad l_n(\theta) - l_n(\theta_0) + n\delta \longrightarrow +\infty \quad P_{\theta_0}^\infty \text{ a.s.}$$

So

$$\int_{S^\varepsilon} \liminf e^{l_n(\theta)-l_n(\theta_0)+n\delta}\, d\pi(\theta) = \int_{S^\varepsilon} +\infty\, d\pi(\theta) = +\infty \quad \text{since } \pi(S^\varepsilon) > 0.$$

So by using ① and ② we obtain:

$$\liminf D_n e^{\delta n} = +\infty \quad P_{\theta_0} \text{ a.s.}$$

i.e. $P_{\theta_0}^\infty\big(\lim D_n e^{\delta n} = +\infty\big) = 1$.

And since $\big(D_n e^{\delta n} \xrightarrow[n\to\infty]{} +\infty\big) \implies \big(\exists n_0,\ \forall n \ge n_0 : D_n > e^{-\delta n}\big)$, we got

$$\mathbb{1}_{D_n \le e^{-n\delta}} \xrightarrow{P_{\theta_0}^\infty \text{ a.s.}} 0. \qquad (\star)$$

If (i) holds, let's show that $\pi(A_n \mid X^{(n)}) \xrightarrow{P_{\theta_0}^\infty \text{ a.s.}} 0$.

$$\begin{aligned}
\pi(A_n \mid X^{(n)}) &= \phi_n\, \pi(A_n \mid X_n) + (1-\phi_n)\, \pi(A_n \mid X_n) \\
&= \phi_n \underbrace{\pi(A_n \mid X_n)}_{\leq 1} + \mathbb{1}_{D_n \leq e^{-n\delta}} (1-\phi_n)\, \pi(A_n \mid X_n) \\
&\qquad + \mathbb{1}_{D_n > e^{-n\delta}} (1-\phi_n)\, \pi(A_n \mid X_n) \\
&\leq \phi_n + \mathbb{1}_{D_n \leq e^{-n\delta}} + \mathbb{1}_{D_n > e^{-n\delta}} (1-\phi_n)\, \frac{\int_{A_n} e^{l_n(\theta) - l_n(\theta_0)}\, d\pi(\theta)}{D_n} \\
&\overset{(\star)}{\leq} \phi_n + \mathbb{1}_{D_n \leq e^{-n\delta}} + \mathbb{1}_{D_n > e^{-n\delta}} (1-\phi_n)\, e^{n\delta} \int_{A_n} e^{l_n(\theta) - l_n(\theta_0)}\, d\pi(\theta)
\end{aligned}$$

So the central inequality is:

$$\pi(A_n \mid X^{(n)}) \leq \underbrace{\phi_n}_{\substack{\xrightarrow{\text{a.s.}} 0 \\ \text{by hypothesis}}} + \underbrace{\mathbb{1}_{D_n \leq e^{-n\delta}}}_{\substack{\longrightarrow 0 \\ \text{by } (\star)}} + \underbrace{\mathbb{1}_{D_n > e^{-n\delta}}\, e^{n\delta} (1-\phi_n)\, I_n}_{\xrightarrow{?} 0}$$

with $I_n := \int_{A_n} e^{l_n(\theta) - l_n(\theta_0)}\, d\pi(\theta)$.

Control of I_n, in order to use Markov

$$\begin{aligned}
\mathbb{E}_{\theta_0}^{(n)}[I_n] &= \mathbb{E}_{\theta_0}^{(n)}\Big[\big(1 - \phi_n(X^{(n)})\big) \int_{A_n} e^{l_n(\theta) - l_n(\theta_0)}\, d\pi(\theta) \Big] \\
&= \int_{\mathcal{X}^n} \int_{A_n} \big(1 - \phi_n(x^{(n)})\big) \frac{f_\theta^{(n)}(x^{(n)})}{\cancel{f_{\theta_0}^{(n)}(x^{(n)})}}\, d\pi(\theta)\, \cancel{f_{\theta_0}^{(n)}(x^{(n)})}\, dx^{(n)} \\
&\overset{\text{Fubini}}{=} \int_{A_n} \mathbb{E}_{\substack{\theta \\ \text{not } \theta_0}}^{(n)} \big[1 - \phi_n(X^{(n)})\big]\, d\pi(\theta) \overset{\text{(i)}}{\leq} e^{-Cn}
\end{aligned}$$

Does $(1-\phi_n)e^{n\delta} I_n \xrightarrow{P_{\theta_0}\ \textbf{a.s.}} 0$?

For $\delta < C/4$:

$$\begin{aligned}
P_{\theta_0}\Big((1-\phi_n) e^{n\delta} I_n > e^{-\frac{Cn}{4}} \Big) &\overset{\text{Markov}}{\leq} e^{n(\frac{C}{4} + \delta)}\, \mathbb{E}_{\theta_0}^{(n)}\big[\underbrace{(1-\phi_n)}_{\leq 1} I_n \big] \\
&\overset{\delta < C/4}{\leq} e^{\frac{nC}{2}}\, \mathbb{E}_{\theta_0}^{(n)}[I_n] \\
&\leq e^{\frac{nC}{2}} e^{-Cn} = e^{-\frac{Cn}{2}} \longrightarrow 0.
\end{aligned}$$

Since $P_{\theta_0}(\dots) \xrightarrow[n\to\infty]{} 0$ exponentially fast with n, by Borel–Cantelli we got $(1-\phi_n)e^{n\delta} I_n \xrightarrow{\text{a.s.}} 0$.

So $\pi(A_n \mid X^{(n)}) \xrightarrow{P_{\theta_0}\ \text{a.s.}} 0$.

If (ii) holds, let's show that $\pi(A_n \mid X^{(n)}) \xrightarrow{P_{\theta_0}^{\infty}\ \text{a.s.}} 0$.

We can show $\phi_n : x^{(n)} \longmapsto 0$ verifies the (i) hypothesis:

- $\phi_n \xrightarrow{P_{\theta_0} \text{ a.s.}} 0$ (trivial)
- $\displaystyle\int_{A_n} \mathbb{E}_\theta^{(n)}[1-\phi_n]\, d\pi(\theta) = \int_{A_n} 1\, d\pi(\theta) = \pi(A_n) \leq e^{-Cn}$

So we got the conclusion by applying (i).

□

3.3 Applications of Schwartz Theorem

3.3.1 The L^1 Metric

Remark (L^1 distance and total variation): TO COMPLETE

Theorem 3.4

Let $\Theta_n \subset \Theta$ be such that

$$\log N\Big(\frac{\varepsilon}{4}, \mathcal{P}_{\Theta_n}, \|\cdot\|_1\Big) \leq \frac{n\varepsilon^2}{16},$$

where $\mathcal{P}_{\Theta_n} := \{f_\theta, \theta \in \Theta_n\}$ and $N(\eta, \mathcal{F}, \|\cdot\|_1)$ is the covering number of $\mathcal{F}$, i.e. the minimal number of $\|\cdot\|_1$-balls of radius η needed to cover $\mathcal{F}$; then $\exists\, \phi_n$ s.t. $\phi_n \longrightarrow 0$ P_{θ_0} a.s. and

$$\sup_{\substack{\theta \in \Theta_n \\ d(\theta,\theta_0)>\varepsilon}} \mathbb{E}_\theta(1-\phi_n) \leq e^{-nC(\varepsilon)} \quad \text{for some } C(\varepsilon) > 0.$$

Lemma 3.5: Posterior consistency

Assume that $\pi(\{\theta;\ \mathrm{KL}(\theta_0;\theta) < \varepsilon\}) > 0$ for all $\varepsilon > 0$ (hypothesis of Schwartz theorem) and that, for every $\varepsilon > 0$, there exist $\Theta_n \subset \Theta$ satisfying the entropy condition of Theorem 3.4 and $C > 0$ such that

$$\pi(\Theta_n^c) \leq e^{-Cn}.$$

Then the posterior is a.s. consistent at θ_0:

$$\forall \varepsilon > 0, \quad \pi\big(\{\theta : d(\theta,\theta_0) > \varepsilon\} \,\big|\, X^{(n)}\big) \xrightarrow[n\to\infty]{} 0 \quad P_{\theta_0}^\infty \text{ a.s.}$$

Proof of Lemma 3.5

Let $\varepsilon > 0$ and $A_n := \Theta_n \cap \{\theta : d(\theta,\theta_0) > \varepsilon\}$. Since

$$\{\theta : d(\theta,\theta_0) > \varepsilon\} \subset A_n \cup \Theta_n^c,$$

we have

$$\pi\big(\{\theta : d(\theta,\theta_0) > \varepsilon\} \,\big|\, X^{(n)}\big) \leq \pi(A_n \mid X^{(n)}) + \pi(\Theta_n^c \mid X^{(n)}).$$

- $\underline{A_n}$: by Theorem 3.4, $\exists\, \phi_n \longrightarrow 0$ $P_{\theta_0}^\infty$ a.s. such that

$$\int_{A_n} \mathbb{E}_\theta^{(n)}(1-\phi_n)\, d\pi(\theta) \leq \pi(A_n) \sup_{\theta \in A_n} \mathbb{E}_\theta^{(n)}(1-\phi_n) \leq e^{-nC(\varepsilon)},$$

so by condition (i) of Schwartz theorem, $\pi(A_n \mid X^{(n)}) \longrightarrow 0$ $P_{\theta_0}^\infty$ a.s.

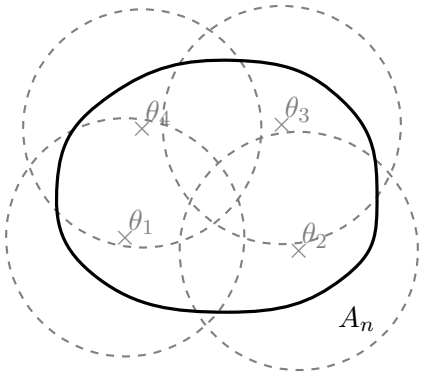
- Θ_n^c: since $\pi(\Theta_n^c) \leq e^{-Cn}$, condition (ii) of Schwartz theorem gives $\pi(\Theta_n^c \mid X^{(n)}) \longrightarrow 0$ $\overline{P_{\theta_0}^\infty}$ a.s.

Hence $\pi(\{\theta : d(\theta, \theta_0) > \varepsilon\} \mid X^{(n)}) \longrightarrow 0$ $P_{\theta_0}^\infty$ a.s. for every $\varepsilon > 0$, which is the consistency at θ_0. □

Remark (testing $H_0 : \theta = \theta_0$ against $H_1 : d(\theta, \theta_0) > \varepsilon$): TO COMPLETE

Proof Intuition for Theorem 3.4

Let $A_n := \Theta_n \cap \{\|f_\theta - f_{\theta_0}\|_1 > \varepsilon\}$. The proof consist in covering A_n with a certain number of balls centered in θ_i with $i \in \{1, \dots, N(\varepsilon/4; A_n; \|\cdot\|_1)\}$, then on each of these balls we construct a local test, ϕ_{n,θ_i}, then by taking $\phi_n := \max_i \phi_{n,i}$, we got the good ϕ_n.



Proof of Theorem 3.4

Construction and expectation bound on local tests.

Let $\theta_1 \in \underbrace{A_n := \Theta_n \cap \{\|f_\theta - f_{\theta_0}\| > \varepsilon\}}_{\text{will be used for Schwartz thm}}$,

$$\phi_{n,\theta_1} := \mathbb{1}_{\frac{1}{n}\left(\sum \mathbb{1}_{R(\theta_1)}(X_i)\right) - P_{\theta_0}(R(\theta_1)) > \frac{\Delta_n(\theta_1)}{n}}$$

with $R(\theta_1) := \{x : f_{\theta_1}(x) > f_{\theta_0}(x)\}$ ($A(\theta_1)$ in Rousseau's notes).

Intuition behind ϕ_{n,θ_1}

If $\theta_1 \neq \theta_0$, by the way $R(\theta_1)$ is defined, the proba under P_{θ_1} for falling in $R(\theta_1)$ is much larger than the proba under P_{θ_0} to fall in $R(\theta_1)$. So

$$\sum \mathbb{1}_{R(\theta_1)}(X_i) - nP_{\theta_0}(R(\theta_1)) \longrightarrow \begin{cases} \infty & \text{under } P_{\theta_1} \text{ by LLN} \\ 0 & \text{under } P_{\theta_0} \end{cases}$$

i.e. $\phi_{n,\theta_1} \xrightarrow{P_{\theta_1}} 1$, $\phi_{n,\theta_1} \xrightarrow{P_{\theta_0}} 0$

Control of type I error (i.e. $\mathbb{E}_{\theta_0}^n[\phi_{n,\theta_1}]$)

$$\mathbb{E}^n_{\theta_0}[\phi_{n,\theta_1}] = P^n_{\theta_0}\Big(\sum \mathbb{1}_{R(\theta_1)}(X_i) - nP_{\theta_0}(R(\theta_1)) > \Delta_n(\theta_1)\Big)$$

Hoeffding inequality: if $Y_1, \dots, Y_n$ are $\perp\!\!\!\perp$ r.v. in $[a_i, b_i]$ a.s. $\forall i \leq n$ then

$$P\Big(\sum (Y_i - \mathbb{E}[Y_i]) \geq t\Big) \leq \exp\Big(-\frac{2t^2}{\sum (b_i - a_i)^2}\Big)$$

We use Hoeffding with

$$Y_i = \mathbb{1}_{R(\theta_1)}(X_i) - P_{\theta_0}(R(\theta_1)) \in \big[-P_{\theta_0}(R(\theta_1));\ 1 - P_{\theta_0}(R(\theta_1))\big]$$

and $t = \Delta_n > 0$, so we got:

$$\mathbb{E}^n_{\theta_0}[\phi_{n,\theta_1}] \leq \exp\big(-2\Delta_n^2(\theta_1)/n\big)$$

Control of type II error (i.e. $\sup_\theta \mathbb{E}^n_\theta[1 - \phi_{n,\theta_1}]$)

Let θ be s.t. $\|f_\theta - f_{\theta_1}\|_1 \leq \delta$ (we will set δ later).

$$\begin{aligned}
\mathbb{E}^n_\theta[1 - \phi_{n,\theta_1}] &= P^n_\theta\Big(\sum_{i=1}^n \mathbb{1}_{R(\theta_1)}(X_i) - nP_{\theta_0}(R(\theta_1)) < \Delta_n(\theta_1)\Big) \\
&\qquad \downarrow + nP_{\theta_0}(R(\theta_1)) - nP_\theta(R(\theta_1)) \\
&= P^n_\theta\Big(\sum_{i=1}^n \mathbb{1}_{R(\theta_1)}(X_i) - nP_\theta(R(\theta_1)) \\
&\qquad\qquad < \Delta_n(\theta_1) + n\big(P_{\theta_0}(R(\theta_1)) - P_\theta(R(\theta_1))\big)\Big)
\end{aligned}$$

(Reversed) Hoeffding inequality (Proof: take $Y_i \leftarrow -Y_i$ in Hoeffding)

$[a_i, b_i]$ a.s. $\forall i \leq n$ then $\forall t > 0$

$$P\Big(\sum (Y_i - \mathbb{E}[Y_i]) < -t\Big) \leq \exp\Big(-\frac{2t^2}{\sum (b_i - a_i)^2}\Big)$$

We want to use Hoeffding inequality with

$$Y_i = \mathbb{1}_{R(\theta_1)}(X_i) - P_\theta(R(\theta_1)) \in \big[-P_\theta(R(\theta_1));\ 1 - P_\theta(R(\theta_1))\big]$$

and $-t = \Delta_n(\theta_1) + n\big(P_{\theta_0}(R(\theta_1)) - P_\theta(R(\theta_1))\big)$, so we need to have $t > 0$, i.e.

$$\begin{gathered}
\Delta_n(\theta_1) + n\big(P_{\theta_0}(R) - P_\theta(R)\big) < 0 \\
\text{i.e.} \quad \Delta_n(\theta_1) < n\big(P_\theta(R) - P_{\theta_0}(R)\big)
\end{gathered}$$

Since $\Delta_n(\theta_1) > 0$, we need to have $P_\theta(R) - P_{\theta_0}(R) > 0$.
Lower bound (not depending on θ) of $P_\theta(R) - P_{\theta_0}(R)$

$(R := R(\theta_1))$

$$\begin{aligned} P_\theta(R) - P_{\theta_0}(R) &= P_{\theta_1}(R) - P_{\theta_0}(R) + P_\theta(R) - P_{\theta_1}(R) \\ &= \underbrace{\int_R (f_{\theta_1} - f_{\theta_0})\, d\mu}_{=\frac{1}{2}\|f_{\theta_0} - f_{\theta_1}\|_1} + \underbrace{\int_R (f_\theta - f_{\theta_1})\, d\mu}_{|\cdot| \le \frac{1}{2}\|f_{\theta_1} - f_\theta\|_1 \le \frac{\delta}{2} \text{ (cond. on } \theta)} \\ &\overset{①}{\ge} \frac{1}{2}\|f_{\theta_0} - f_{\theta_1}\|_1 - \frac{\delta}{2} \quad > 0 \leftarrow \text{what we want} \end{aligned}$$

So in order to have $t > 0$, one can take

$$\begin{cases} \delta < \|f_{\theta_0} - f_{\theta_1}\| \\ \Delta_n < n\big(\frac{1}{2}\|f_{\theta_0} - f_{\theta_1}\|_1 - \frac{\delta}{2}\big) \end{cases}$$

So the Hoeffding inequality with such δ and Δ_n gives:

$$\mathbb{E}_\theta^n[1 - \phi_{n,\theta_1}] \le \exp\left(- \frac{2\Big[\Delta_n(\theta_1) - n\big(P_\theta(R) - P_{\theta_0}(R)\big)\Big]^2}{n} \right).$$

Since

$$\Delta_n(\theta_1) - n\big(P_\theta(R) - P_{\theta_0}(R)\big) \overset{①}{\le} \Delta_n - n\Big(\frac{1}{2}\|f_{\theta_0} - f_{\theta_1}\|_1 - \frac{\delta}{2}\Big) \le 0$$

and $\exp(-\cdot^2)$ is increasing on $\mathbb{R}_-$, we got:

$$\begin{aligned} \mathbb{E}_\theta^n[1 - \phi_{n,\theta_1}] &\le \exp\left(- \frac{2\Big(\Delta_n - n\big(\frac{1}{2}\|f_{\theta_0} - f_{\theta_1}\|_1 - \frac{\delta}{2}\big)\Big)^2}{n} \right) \\ &= \exp\Big(- 2n\Big(\frac{\Delta_n}{n} - \frac{1}{2}\|f_{\theta_0} - f_{\theta_1}\|_1 + \frac{\delta}{2}\Big)^2 \Big) \end{aligned}$$

We want $\dfrac{\Delta_n}{n} - \dfrac{1}{2}\|f_{\theta_0} - f_{\theta_1}\|_1 + \dfrac{\delta}{2} < c < 0$.
We want to find a solution to the following system in δ and Δ_n that don't depend on θ_1, so that we can have a uniform bound:

$$\begin{cases} \dfrac{\Delta_n}{n} - \dfrac{1}{2}\|f_{\theta_0} - f_{\theta_1}\| + \dfrac{\delta}{2} < c < 0 & (1) \\ \delta \le \underbrace{\|f_{\theta_0} - f_{\theta_1}\|}_{>\varepsilon} & (2) \\ \Delta_n < n\big(\frac{1}{2}\|f_{\theta_0} - f_{\theta_1}\|_1 - \frac{\delta}{2}\big) & (3)\ (\Leftarrow (1)) \\ \Delta_n > 0 & (4) \\ \Delta_n^2 > n^2 C_2, \quad \exists\, C_2 > 0 \text{ (for type I control)} & (5) \end{cases}$$

For $\delta = \dfrac{\varepsilon}{4}$, $\Delta_n = n\dfrac{\varepsilon}{4}$:

$$\Longleftarrow \begin{cases} (1) \quad \dfrac{\varepsilon}{4} - \dfrac{1}{2}\|f_{\theta_0} - f_{\theta_1}\| + \dfrac{\varepsilon}{8} \underset{\|f_{\theta_0} - f_{\theta_1}\| > \varepsilon}{<} -\dfrac{\|f_{\theta_0} - f_{\theta_1}\|}{4} + \dfrac{\varepsilon}{8} < 0 \quad \checkmark \\ (3),\ (2),\ (4),\ (5) \quad \checkmark \end{cases}$$

So we get $\forall \theta \in B_{\|\cdot\|_1}(\theta_1, \delta)$,

$$\mathbb{E}_\theta^{(n)}[1 - \phi_{n,\theta_1}] \leq \exp(-2nc)$$

so

$$\sup_{\theta \text{ s.t. } \|f_\theta - f_{\theta_1}\|_1 < \delta} \mathbb{E}_\theta^{(n)}[1 - \phi_{n,\theta_1}] \leq \exp(-2nc)$$

and

$$\mathbb{E}_{\theta_0}^{(n)}[\phi_{n,\theta_1}] \leq \exp(-2nC_2)$$

Global control of type I and II errors

Let $(\theta_j)_{j=1}^{N_n}$ be a covering of A_n by balls of radius $\delta = \frac{\varepsilon}{4}$ (with N_n covering number $< \infty$).
Let $\phi_n := \max_{j \leq N_n} \phi_{n,\theta_j}$.

Type I control:

then

$$\mathbb{E}_{\theta_0}^{(n)}[\phi_n] \leq \sum_{j=1}^{N_n} \mathbb{E}_{\theta_0}^{(n)}[\phi_{n,\theta_j}] \leq N_n \exp(-2nC_2)$$

So by Borel–Cantelli $\phi_n \xrightarrow[n\to\infty]{P_{\theta_0} \text{ a.s.}} 0$.

Type II control:

$$\begin{aligned}\sup_{\theta \in A_n} \mathbb{E}_\theta[1 - \phi_n] &= \sup_j \sup_{\theta : \|f_\theta - f_{\theta_j}\|_1 < \delta} \mathbb{E}_\theta[1 - \phi_n] \\ &\leq \sup_j \exp(-2nc)\end{aligned}$$

□

3.3.2 Hellinger Tests

Theorem 3.6: Existence of Hellinger Tests

TO COMPLETE

Corollary 3.7: Tests for the Hellinger Distance

TO COMPLETE

Proof

TO COMPLETE □

3.4 Parametric Models

TO COMPLETE

Theorem 3.8: Consistency in Parametric Models

TO COMPLETE

Proof

TO COMPLETE □

Example 3.9: Normal Location Model

TO COMPLETE

Example 3.10: Uniform Model $\mathcal{U}([0, \theta])$

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4 Posterior Convergence Rate

4.1 Definitions and Theorem

Definition 4.1: Posterior Contraction Rate

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Intuition

TO COMPLETE

Theorem 4.2: Posterior Contraction Rate

TO COMPLETE

Remark (i.i.d. observations): **TO COMPLETE**

Lemma 4.3: Lower Bound on D_n

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Proof

TO COMPLETE □

Proof of Theorem 4.2

TO COMPLETE □

Proposition 4.4: Key Inequality

TO COMPLETE

Corollary 4.5: I.I.D. Case with the Hellinger or L^1 Distance

TO COMPLETE

Proof

TO COMPLETE □

4.2 Consequence on the Convergence of the Posterior Mean

TO COMPLETE
Remark (unbounded parameter space): TO COMPLETE

Example 4.6: Contraction Rate in the Normal Location Model

TO COMPLETE

Exercise (contraction rate in linear regression): TO COMPLETE
Exercise (nonparametric regression): TO COMPLETE